MACHINE LEARNING

LESSON 05:

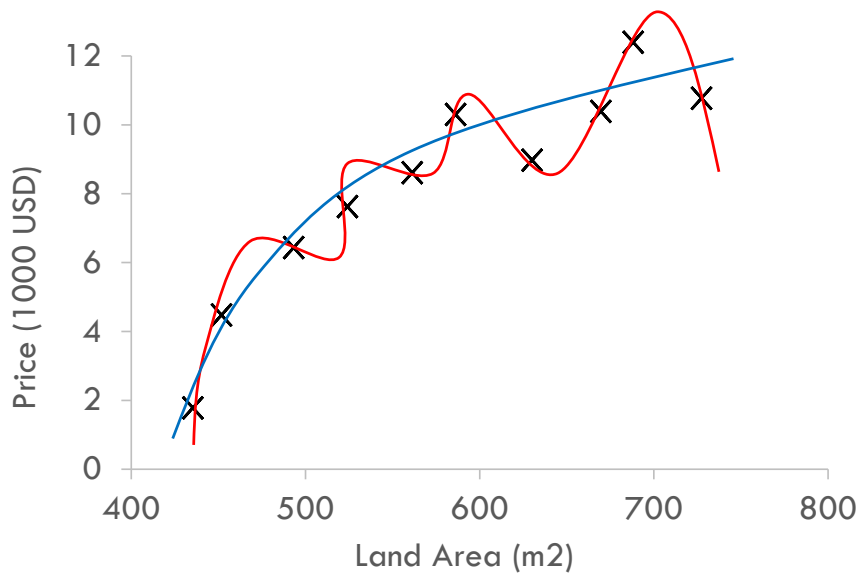
Model Selection and Evaluation

Prepared by Dr. Dona Valy

Inspired by and adapted from the Coursera Machine Learning course by Andrew Ng (coursera.org/learn/machine-learning)

HYPOTHESIS EVALUATION

- Overfitting Problem



It fails to generalize to new examples not in the training set

Dataset with many input features?

HYPOTHESIS EVALUATION

- Dataset

Size in m^2 (x)	Price in 1000 USD (y)
90	7.1
130	10.9
210	19.2
300	28
350	32.8
420	39.9
480	46.1
530	51
640	62.2
710	68.9

Training Set

$$(x^{(1)}, y^{(1)})$$
$$(x^{(2)}, y^{(2)})$$
$$\dots$$
$$(x^{(m)}, y^{(m)})$$

Test Set

$$(x_{test}^{(1)}, y_{test}^{(1)})$$
$$(x_{test}^{(2)}, y_{test}^{(2)})$$
$$\dots$$
$$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$$

HYPOTHESIS EVALUATION

- Training/testing procedure
  - Learn parameter Θ from training data (by minimizing error/cost $J(\Theta)$)
    - $J(\Theta) = \frac{1}{m}\sum_{i=1}^{m} C\left(h\left(x^{(i)}\right), y^{(i)}\right)$
  - Compute test set error:
    - $J_{test}(\Theta) = \frac{1}{m_{test}}\sum_{i=1}^{m_{test}} C\left(h\left(x_{test}^{(i)}\right), y_{test}^{(i)}\right)$
  - In the case of logistic regression/classification problem, a misclassification error can be computed:
    - $err(h(x), y) = 1$ if it x is classified incorrectly or 0 otherwise

$$\Rightarrow test\ error = \frac{1}{m_{test}}\sum_{i=1}^{m_{test}} err\left(h\left(x_{test}^{(i)}\right), y_{test}^{(i)}\right)$$

MODEL SELECTION

- Example: we build a neural network with l hidden layers each layer with k units
  - Model 1: $h_1(x), l = 1 \Rightarrow \Theta_1 \Rightarrow J_{test}(\Theta_1)$
  - Model 2: $h_2(x), l = 2 \Rightarrow \Theta_2 \Rightarrow J_{test}(\Theta_2)$

  - Model 10: $h_{10}(x), l = 10 \Rightarrow \Theta_{10} \Rightarrow J_{test}(\Theta_3)$

- How well does each model generalize?
  - Compute test set error $J_{test}(\Theta_l)$
  - **Problem:** the extra parameter (hyperparameter) l might fit to the test set

MODEL SELECTION

- Dataset

Size in m^2 (x)	Price in 1000 USD (y)
90	7.1
130	10.9
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300	28
350	32.8
420	39.9
480	46.1
530	51
640	62.2
710	68.9

Training Set

$$\left(x^{(1)}, y^{(1)}\right)$$
$$\left(x^{(2)}, y^{(2)}\right)$$
$$\dots$$
$$\left(x^{(m)}, y^{(m)}\right)$$

Validation Set

$$\left(x_v^{(1)}, y_v^{(1)}\right)$$
$$\left(x_v^{(2)}, y_v^{(2)}\right)$$
$$\dots$$
$$\left(x_v^{(m_v)}, y_v^{(m_v)}\right)$$

Test Set

$$\left(x_{test}^{(1)}, y_{test}^{(1)}\right)$$
$$\left(x_{test}^{(2)}, y_{test}^{(2)}\right)$$
$$\dots$$
$$\left(x_{test}^{(m_{test})}, y_{test}^{(m_{test})}\right)$$

MODEL SELECTION

- Train/validation/test error
  - Training error: $J(\Theta) = \frac{1}{m}\sum_{i=1}^{m} C\left(h\left(x^{(i)}\right), y^{(i)}\right)$
  - Validation error: $J_v(\Theta) = \frac{1}{m_v}\sum_{i=1}^{m_v} C\left(h\left(x_v^{(i)}\right), y_v^{(i)}\right)$
  - Test error: $J_{test}(\Theta) = \frac{1}{m_{test}}\sum_{i=1}^{m_{test}} C\left(h\left(x_{test}^{(i)}\right), y_{test}^{(i)}\right)$

- For each model, we minimize $J(\Theta) \Rightarrow J_v(\Theta)$
  - Choose the best model using the validation set
  - Estimate generalization error on test set $J_{test}(\Theta)$

ERROR METRICS FOR SKEWED CLASSES

- Example: Cancer classification
  - Train a binary classification model $h(x)$ ($y = 1$ if the patient has cancer or 0 otherwise)
  - Find out that you got 1% error on test set; however, only 0.5% of the patients have cancer in the test set
  - What if we compare our model with another model which always predict that the patient does not have cancer for any input x?

PRECISION AND RECALL

- Precision: of all patients where we predict $y = 1$, what fraction actually has cancer?

$$\frac{TP}{TP + FP}$$

- Recall: of all patients that actually have cancer, what fraction is correctly detected as having cancer?

$$\frac{TP}{TP + FN}$$

		Actual Output	
		1	0
Predicted Class	1	True Positive (TP)	False Positive (FP)
	0	False Negative (FN)	True Negative (TN)

F SCORE

- How to compare precision/recall numbers?

	Precision (P)	Recall (R)
Model 1	0.9	0.2
Model 2	0.4	0.7
Model 3	0.1	1

Average
0.55
0.55
0.55

F Score
0.33
0.51
0.18

- F Score: $2\frac{PR}{P+R}$

EXERCISE 5.1

Look back at the logistic regression model corresponding to $h_2(x)$ in Exercise 3.1.2. Divide the dataset into 60% and 40% for training and test set respectively. Retrain the model on the training set and evaluate it on the test set using precision, recall, and F score metrics.

EXERCISE 5.2

Given a dataset about chemical measurements of wine*, build a neural network as a classifier to predict the quality (Medium, Good, or Excellent) of wine. Divide the dataset into 3 subsets: training (60%), validation (10%), and test (30%). Use the validation set to choose the number of hidden layers (between 2 and 5 hidden layers with 50 units each) for your neural network. Finally compute the error rate of the model that you select on the test set.

* Dataset from https://machinelearningmastery.com/standard-machine-learning-datasets

EXERCISE 5.3

Given a dataset containing 10 classes of images of handwritten Khmer characters 'kh_char_10_class.zip', divide the training set into two subsets: new training set (80%) and validation set (20%). Use the validation set to find the best possible neural network model evaluated on the test set.